

Arjuna JEE 2025

Mathematics

Binomial Theorem

DPP: 3

Q1 The coefficient of middle term in the expansion of $(1+x)^n$ when n is even, is

- (A) $\frac{1 \cdot 3 \cdot 5 \dots n}{2 \cdot 4 \cdot 6 \dots (n+1)} \cdot 2^n$
 (B) $\frac{1 \cdot 3 \cdot 5 \dots (n-1)}{2 \cdot 4 \cdot 6 \dots n} \cdot 2^n$
 (C) $\frac{2 \cdot 4 \cdot 6 \dots n}{1 \cdot 3 \cdot 5 \dots (n-1)} \cdot 2^n$
 (D) None of these

Q2 The coefficient of the middle term in the binomial expansion in powers of x of $(1+\alpha x)^4$ and of $(1-\alpha x)^6$ is the same, if α is equal to

- (A) $-5/3$ (B) $10/3$
 (C) $-3/10$ (D) $3/5$

Q3 The greatest term in the expansion of $(1+3x)^{54}$ where $x = 1/3$ is

- (A) T_{28} (B) T_{25}
 (C) T_{26} (D) T_{24}

Q4 The 4^{th} term in the expansion of $(2 + \frac{3x}{8})^{10}$ has the maximum numerical value then the value of x can be in the interval

- (A) $[-\frac{64}{21}, -2] \cup [2, \frac{64}{21}]$
 (B) $(-\infty, -\frac{64}{21})$
 (C) $(\frac{64}{21}, \infty)$
 (D) None

Q5 The greatest term in the expansion of $(1+3x)^{54}$, where $x = 1/3$ is

- (A) T_{28} (B) T_{25}
 (C) T_{26} (D) T_{24}

Q6

The greatest term in the expansion of $(3-5x)^{15}$, when $x = \frac{1}{5}$, is

- (A) 6^{th}
 (B) 5^{th}
 (C) $5^{\text{th}} \& 6^{\text{th}}$
 (D) $4^{\text{th}} \& 5^{\text{th}}$

Q7 Find the remainder when the number 9^{100} is divided by 8.

Q8 If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

Q9 If the remainder when x is divided by 4 is 3, then the remainder when $(2020+x)^{2022}$ is divided by 8 is _____.

Q10 If $\{p\}$ denotes the fractional part of the number p , then $\left\{\frac{3^{200}}{8}\right\}$, is equal to:

- (A) $\frac{5}{8}$
 (B) $\frac{7}{8}$
 (C) $\frac{1}{8}$
 (D) $\frac{3}{8}$

Q11 What is the remainder when 5^{99} is divided by 13

Q12 The last two digits of 3^{400} are

- (A) 01 (B) 29
 (C) 43 (D) 81

Q13 The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9, is

- (A) 7 (B) 8
 (C) 0 (D) 2



- Q14** If $2^{2020} + 2021$ is divided by 9, then the remainder obtained is
(A) 0 (B) 1
(C) 3 (D) 7
- Q15** If $n \in \mathbb{N}$, then remainder when $(37)^{n+2} + 16^{n+1} + 30^n$ is divided by 7 is
- Q16** The remainder when 7^{103} is divided by 25, is
(A) 15 (B) 16
(C) 17 (D) 18
- Q17** The last three digits of the number $N = 7^{100} - 3^{100}$ are
(A) 100 (B) 300
(C) 500 (D) 000



Answer Key

Q1 (B)

Q2 (C)

Q3 (A)

Q4 (A)

Q5 (A)

Q6 (D)

Q7 1

Q8 4

Q9 1

Q10 (C)

Q11 8

Q12 (A)

Q13 (D)

Q14 (C)

Q15 0

Q16 (D)

Q17 (D)

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